

PS200C Causal Inference, Spring 2026

Introduction

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...what do we really learn from such observational evidence?

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The trial was **halted early**.

- Physicians reversed course overnight
- Millions of women stopped HRT
- Even now people take guidance from that trial.

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- The “timing hypothesis” is now widely accepted, and guidelines have largely come back around.

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Lesson: Even a large RCT doesn't necessarily tell you what you want.

- Design details (the treatment formulation, who is enrolled), stopping rules, interpretation can invite errors
- An experiment answers a causal question, but *which* causal question depends on details you cannot ignore.

More examples to keep in mind

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- **Microfinance:** Celebrated as a path out of poverty based on observational comparisons of borrowers vs. non-borrowers. Six major RCTs found much more modest effects—helpful in some ways, but not transformative.
- **Smoking and lung cancer:** We will never have an RCT. Yet the causal conclusion is among the most trusted in all of science. Why? Not because of the volume of data, but because researchers could formalize *what it would take* for the result to be wrong—and the required confounding was implausible. (We will return to this when we cover sensitivity analysis.)

Modes of Statistical Inference

1. Descriptive Inference: summarizing and exploring data
 - means, correlations, regression coefficients
 - inferring “ideal points” from roll-call votes
 - inferring “topics” from texts and speeches
 - Note: the statistical inference you are used to—hypothesis tests, p -values, confidence intervals—is *descriptive* inference. It furthers the description of how variables look or move together. It does not, by itself, get you any closer to causal.

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 - predicting future state failures from past failures and covariates
 - predicting vote share from a survey of likely-voters
3. **Causal Inference**: engages *counterfactuals*
 - inferring whether the lack of war among democracies can be attributed to regime type
 - determining whether a medical treatment improved outcomes.
 - generally, estimating effect of some policy or institution or treatment on some outcome...

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- **Yes**: it is a different way of thinking that may be awkward and magical sounding at first!
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In fact, I hope the course helps to convince you that technical sophistication of methods often does not solve core problems...it's not about the latest estimator, machine learning, or more complicated standard errors.

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Stylized illustration:

- you have data on X and Y
- you want to learn some effect of X on Y
- you compute something with data, say, $\text{coef}(Y \sim X)$
- trust me (for now): *without further assumption*, any causal effect can produce any coefficient...and thus any coefficient you get is equally suggestive of any causal effect.

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So what *does* it take? That's what this course is about.

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- Credibility does NOT derive from “what method you used.” There are not “causal methods”! You can't say “it's causal because I did xyz.”
- This is one of the hardest things for many students to absorb, in part because we are surrounded by mistakes and false advertising.

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- Randomized experiments don’t make causal reasoning unnecessary—they make it *more* necessary. What **mechanism** produced the total effect? How do you handle **attrition** (which is post-treatment)? How do you choose covariates for **precision** without introducing bias? These all require understanding causal structure.

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- Randomized experiments don’t make causal reasoning unnecessary—they make it *more* necessary. What **mechanism** produced the total effect? How do you handle **attrition** (which is post-treatment)? How do you choose covariates for **precision** without introducing bias? These all require understanding causal structure.
- Even **generalization**—asking whether a result applies beyond the study sample—is a causal question about what would happen under different conditions.

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- How can we understand how vulnerable a given estimate might be to threats such as confounding?
- **How can you conduct top quality work that seeks to make causal claims in tricky situations, while accurately characterizing what can and cannot be claimed?**

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- In lecture, the emphasis is on **intuition and assumptions**—truly understanding the (heroic) assumptions required by each identification strategy.
- You are expected and encouraged to **use AI tools** (Claude, ChatGPT, etc.) as tutors. We want you to learn to *think* about causal inference; the tools can help you *do* it.

Topics

#	Topic
1	Introduction and the causal inference problem
2	Potential outcomes and the identification problem
3	Randomized experiments
4	Selection on observables
5	DAGs and structural causal models
6	Sensitivity analysis
7	Difference-in-differences
8	Instrumental variables
9	Regression discontinuity
10	Special topics (time permitting)

Topics do not map exactly to weeks. Detailed readings for each topic will be posted on the course website.

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We will distribute a short **diagnostic self-assessment** (ungraded) to help you identify any gaps. The Resources page has materials keyed to each topic.

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- A more comprehensive and annotated set of resources is on the course website's **Resources** page.

Course Requirements and Grades

- **In-class activities (20%)**

- Short (5–10 min) pen-and-paper exercises at end of most lectures
- Graded on completion and good-faith effort
- Lowest two dropped (accommodates absences)
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- **Final project (50%)**

- Research paper (~10–15 pages) in groups of 1–3
- Presentations in the last week of class
- Due during finals week

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- Be honest about limitations—discuss which assumptions are most vulnerable.
- You may use AI tools freely for coding, data wrangling, and exploring ideas. Include a short section describing how you used AI in the research process.
- What AI *cannot* do for you: choose an identification strategy, evaluate whether its assumptions are credible, or interpret what results do and do not tell you. That is your job.

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- This is not assessed for quality—it normalizes the practice and gives us insight into what topics are causing confusion.

Section

- **Your TA:** Barney Chen — `barneychen@ucla.edu`

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- Section: Thursday 9:00–9:50 AM (Zoom — see course website for link)

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- Barney's office hours: Mondays 5 PM, Bunche 3244 (and by appointment)

Logistics

- **Class:** MW 2:00–3:15 PM, Bunche 2160

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- **Course website:**

<https://chadhazlett.github.io/ps200c-2026/>

Syllabus, slides, resources, project guidelines, section materials.

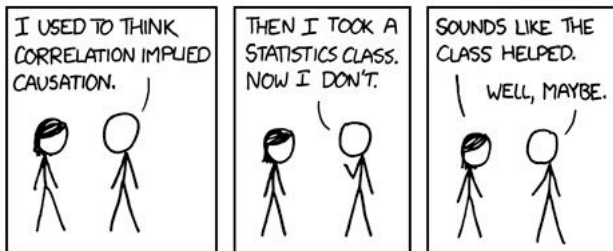
- **Class:** MW 2:00–3:15 PM, Bunche 2160
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- **My office hours:** Wednesday 3:30–4:45 PM and by appointment, Bunche 3264.

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- Come to class, engage, ask questions. If you aren't going to attend regularly, don't take the class.

Some Resources

- 1 CAPS: <https://www.counseling.ucla.edu>. They also have a 24/7 crisis support at 310-825-0768.
- 2 The Bruin Fund offers funding for students with computing needs.
- 3 I'm always happy to help you get any services you need or to figure out what's available.



- Questions?
- Introductions!